

ABG Interpretation Workshop

Faculty Detailed Guide

How this session runs

All 13 cases handed out. 8-minute settling period. Each case = 15-minute unit.

Small Group - 8 min - Group faculty	Plenary - 7 min - Moderator
0-3 min: Team interprets independently 3-6 min: Faculty-mediated discussion 6-8 min: Group synthesises answer	0-3 min: Random draw - one group presents 3-6 min: Case-specific Q and A plus post-intervention gas 6-7 min: Take-home pearl

Timing discipline: hold the 3-3-2 boundary. Resist answering in the first 3 minutes. If the group finishes early, ask a what-if variant.

Case 1 twist: small-group phase works presenting gas only; moderator reveals the 14-hour repeat gas in plenary.

Quick Answer Index - all 13 cases

#	Diagnosis	Moderator take-home
1	DKA	Recalculate the anion gap on every repeat gas - a persistently low HCO_3^- after treatment can mean the acid has changed character.
2	Pyloric stenosis	A urine chloride confirms whether a metabolic alkalosis will respond to chloride repletion. SID tracks the response.
3	Gastroenteritis	A normal-anion-gap acidosis with a normal lactate points to GI bicarbonate loss, not hypoperfusion. SID confirms the chloride-driven mechanism.
4	Suspected IEM	Correct the anion gap for albumin as a reflex, every time - it is most consequential exactly when the raw number looks reassuring. Also check SID.
5	Salicylate poisoning	A near-normal pH never means a normal gas - check PaCO_2 and HCO_3^- against each other before you trust it. SID helps distinguish chloride-driven from unmeasured-anion-driven acidosis.
6	Post-arrest hanging	When PaCO_2 and HCO_3^- are both markedly abnormal in the same direction, name both disorders explicitly - one is not compensating for the other. SID confirms no chloride-driven component.
7	Septic shock + severe anaemia	A normal SpO_2 says nothing about oxygen delivery when haemoglobin is this low - look at ScvO_2 and lactate, not the saturation monitor. SID reveals the chloride-driven component.
8	Methaemoglobinaemia	A high PaO_2 does not guarantee oxygen-carrying capacity - whenever SpO_2 and PaO_2 disagree, request co-oximetry before you do anything else.
9	Two encephalopathic neonates	Neonatal encephalopathy + abnormal gas = metabolic screen; ammonia on ice is critical. SID helps distinguish chloride-driven from unmeasured-anion-driven acidosis.
10	HUS-like	HUS = mixed acidosis; hyperkalaemia and hypocalcaemia are immediate threats. SID, Atot, and unmeasured anions each tell a different part of the story.
11	Hepatitis A encephalopathy	Hepatitis A encephalopathy = respiratory alkalosis + HAGMA; check glucose. Always correct the AG for albumin, and check SID - the SBE may look normal while a HAGMA is masked.
12	Fasting hypoglycaemia	Fasting hypoglycaemia with low ketones = think fatty acid oxidation defect; critical sample before treatment. SID confirms the acidosis was mild and chloride-driven, not the primary problem.
13	Shocked 2-week-old boy	Neonatal shock + NAGMA + hyperkalaemia + hyponatraemia = CAH until proven otherwise. SID confirms the chloride-driven mechanism and tracks response to therapy.

The Unified 10-Step Approach

Step	Tool / key question
1. Clinical history	What does the story alone suggest?
2. Internal validity	$[H^+] = 24 \times PaCO_2 / HCO_3^-$ - does it match pH? Arterial?
3. pH	Acidaemia, alkalaemia, or normal? A normal pH does not mean a normal gas.
4. Primary disorder	Does $PaCO_2$ or HCO_3^- explain the pH direction?
5. Expected compensation	Metabolic acidosis: $PaCO_2 = 1.5 \times HCO_3^- + 8 \pm 2$. Metabolic alkalosis: $PaCO_2 = 0.7 \times HCO_3^- + 21 \pm 2$.
6. Anion gap	$Na^+ - (Cl^- + HCO_3^-)$; add 2.5 per 1 g/dL albumin below 4.0.
7. Delta ratio	$(AG - 12)/(24 - HCO_3^-)$ - only if AG raised.
8. Oxygenation & integration	P/F, OI, CaO_2 , $ScvO_2$, MetHb. Does the story fit the child?
9. SID / Stewart analysis	SID = $Na^+ - Cl^-$ (normal ~36). Low = chloride-driven acidosis. High = chloride-loss alkalosis. Atot = albumin + phosphate. Unmeasured anions = $AG - (lactate + ketones)$.
10. Post-intervention re-evaluation	Has the mechanism changed? Recalculate AG, SID, Atot.

Case 1 - DKA

Small-group phase (8 min) - you run this

Vignette. 10-year-old girl, 2 weeks of polyuria and 3 days of vomiting and abdominal pain. Acidotic breathing present, haemodynamically stable, GCS 11.

Presenting gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.08	14	4	36	132
K	Cl	Albumin	Glucose	b-OHB
4.5	98	4.0	480	5.8

Worked interpretation

Step	Answer
1. Clinical history	2 weeks of polyuria (osmotic diuresis) with vomiting and abdominal pain, Kussmaul breathing, GCS 11 - classical new-onset DKA with early obtundation.
2. Internal validity	$[H^+] = 24 \times 14 / 4 = 84 \text{ nmol/L} \rightarrow \text{pH } 7.08$ - matches. PaO ₂ 36 suggests a venous sample.
3. pH	7.08 - severe acidaemia.
4. Primary disorder	HCO ₃ 4 (low) with acidaemia $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $\text{PaCO}_2 = 1.5 \times 4 + 8 = 14 \pm 2$. Actual 14 - appropriate.
6. Anion gap	$\text{AG} = 132 - (98 + 4) = 30$ (markedly raised). Albumin normal - no correction needed.
7. Delta ratio	$(30 - 12)/(24 - 4) = 0.9 \rightarrow$ pure HAGMA.
8. Oxygenation & integration	Severe HAGMA with maximal appropriate respiratory compensation; AG 30 fully explained by ketoacids (b-OHB 5.8). Severe DKA.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$132 - 98 = 34$	Low-normal - mild chloride-driven component already present.
Atot	Albumin 4.0	Normal.
Unmeasured anions	$30 - 5.8 = 24.2$	Markedly raised - ketoacids dominate.
Mechanism	-	Unmeasured anions, with mild chloride-driven component.

Where groups go wrong

- Calling this respiratory alkalosis from the low PaCO₂ alone.
- Missing that the low PaO₂ (venous sample) is itself a validity teaching point.
- Forgetting to correct the anion gap for albumin as a reflex step.

Your synthesis (final 2 min)

Severe DKA, appropriately compensated, pure HAGMA. Plan: fluid resuscitation, insulin infusion with potassium replacement anticipated.

Plenary phase (7 min) - moderator runs this

Case-specific questions

Q1. Calculate the corrected serum sodium.

Q2. What do the glucose and b-OHB values suggest?

Answers

A1. $\text{Na}^+ + 1.6 \times [(480 - 100)/100] \sim 138$. True hyponatraemia is dilutional from hyperglycaemia.

A2. Hyperglycaemic ketosis driving the gap - severe DKA.

Post-intervention reveal

Intervention. 10 mL/kg NS bolus, insulin infusion, NS + KCl.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.25	26	11	34	142
K	Cl	Albumin	Glucose	b-OHB
3.4	119	4.0	230	0.6

Step	Answer
2. Internal validity	[H ⁺] ~56 nmol/L - internally valid.
3. pH	7.25 - improving.
4. Primary disorder	HCO ₃ 11 -> metabolic acidosis.
5. Expected compensation	Winter's: PaCO ₂ = 1.5 x 11 + 8 = 24.5 +/-2. Actual 26 - appropriate.
6. Anion gap	AG = 142 - (119 + 11) = 12 (normal - HAGMA resolved).
7. Delta ratio	(12 - 12)/(24 - 11) = 0 -> pure NAGMA.
9. SID / Stewart	SID = 142 - 119 = 23 (markedly low - chloride-driven). Unmeasured anions = 12 - 0.6 = 11.4 (normal). Mechanism changed: ketoacids cleared, chloride load now driving.

Post-intervention case-specific answers

Q1. Compare this ABG with the initial ABG. What has changed?

Q2. Why is acidosis persisting despite improvement in glucose and b-OHB?

Q3. Calculate the corrected serum sodium.

A1. pH improved (7.08->7.25); PaCO₂ rose (14->26); HCO₃ rose modestly (4->11); glucose fell (480->230); b-OHB fell (5.8->0.6); Na rose (132->142); K fell (4.5->3.4); Cl rose sharply (98->119); AG normalised (30->12). The acid has changed character.

A2. Ketoacidosis resolved. Persisting low HCO₃ is now hyperchloraemic NAGMA from saline chloride load - expected, not treatment failure.

A3. $142 + 1.6 \times [(230 - 100)/100] \sim 144$.

Moderator take-home (final 1 min)

Recalculate the anion gap on every repeat gas - a persistently low HCO₃ after treatment can mean the acid has changed character.

Case 2 - Pyloric stenosis

Small-group phase (8 min) - you run this

Vignette. 6-week-old first-born boy with 10 days of progressively forceful, non-bilious vomiting after every feed. Sunken eyes, lethargic but arousable, still hungry after vomiting.

Presenting gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.52	48	38	34	132
K	Cl	Albumin	Glucose	Urine Cl
3.0	82	4.0	90	<15

Worked interpretation

Step	Answer
1. Clinical history	Classic for hypertrophic pyloric stenosis: forceful non-bilious vomiting after every feed, still hungry, with dehydration.
2. Internal validity	$[H^+] = 24 \times 48 / 38 = 30 \rightarrow$ pH 7.52 - matches. PaO ₂ 34 suggests venous.
3. pH	7.52 - alkalaemia.
4. Primary disorder	HCO ₃ 38 (raised) with alkalaemia $\rightarrow$ metabolic alkalosis.
5. Expected compensation	$PaCO_2 = 0.7 \times 38 + 21 = 47.6 \pm 2$ (45.6-49.6). Actual 48 - appropriate.
6. Anion gap	$AG = 132 - (82 + 38) = 12$ (normal).
7. Delta ratio	Not applicable - no HAGMA.
8. Oxygenation & integration	Hypochloraemic, hypokalaemic metabolic alkalosis with appropriate respiratory compensation; low urine Cl confirms chloride-responsive process.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$132 - 82 = 50$	Markedly high - chloride-loss alkalosis.
Atot	Albumin 4.0	Normal.
Unmeasured anions	$AG\ 12 - 0 = 12$	Normal.
Mechanism	-	Loss of gastric HCl has removed chloride, raising SID.

Where groups go wrong

- Trying to apply the delta ratio or AG framework to an alkalosis.
- Overlooking K 3.0 as an urgent, actionable finding.

Your synthesis (final 2 min)

A textbook gas: use it to drill the metabolic-alkalosis compensation formula and the urine-chloride distinction.

Plenary phase (7 min) - moderator runs this

Case-specific questions

Q1. What is the significance of the low urine chloride?

Q2. What is the likely clinical diagnosis?

Answers

A1. Urine Cl < 15-20 mmol/L confirms chloride-responsive metabolic alkalosis: kidney appropriately conserving chloride. Alkalosis will correct with chloride-rich fluid. Contrast with chloride-resistant causes (Bartter/Gitelman) where urine Cl is inappropriately high.

A2. Hypertrophic pyloric stenosis - persistent vomiting of gastric HCl and volume depletion produce hypochloraemic, hypokalaemic metabolic alkalosis; secondary hyperaldosteronism drives further renal H⁺ and K⁺ loss.

Post-intervention reveal

Intervention. IV 0.9% saline + KCl, correct dehydration and hypokalaemia, then pyloromyotomy.

pH	PaCO ₂	HCO ₃	Na	K
7.45	40	27	138	4.0
Cl	Albumin	Urine Cl		
100	4.0	>20		

Step	Answer
9. SID / Stewart	SID = 138 - 100 = 38 (normalising - chloride repletion working). Urine Cl now >20 confirms kidney no longer conserving chloride.

Moderator take-home (final 1 min)

A urine chloride confirms whether a metabolic alkalosis will respond to chloride repletion. SID tracks the response.

Case 3 - Gastroenteritis

Small-group phase (8 min) - you run this

Vignette. 8-month-old girl with 2 days of fever and large-volume loose stools. Sunken eyes, delayed skin turgor, prolonged capillary refill time.

Presenting gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.22	27	11	34	129
K	Cl	Albumin	Glucose	Lactate
3.2	112	3.5	85	1

Worked interpretation

Step	Answer
1. Clinical history	Acute gastroenteritis with large-volume diarrhoea and significant dehydration.
2. Internal validity	$[H^+] = 24 \times 27 / 11 = 59 \rightarrow$ pH 7.23 - matches. PaO ₂ 34 suggests venous.
3. pH	7.22 - acidaemia.
4. Primary disorder	HCO ₃ 11 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 11 + 8 = 24.5 \pm 2$ (22.5-26.5). Actual 27 is at upper edge - borderline compensation.
6. Anion gap	$AG = 129 - (112 + 11) = 6$ (low/normal). Corrected $AG = 6 + 2.5 \times (4 - 3.5) = 7.25$ - still normal.
7. Delta ratio	Not applicable - no raised AG.
8. Oxygenation & integration	NAGMA (hyperchloraemic) from stool bicarbonate loss; normal lactate argues against hypoperfusion.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$129 - 112 = 17$	Markedly low - severe chloride-driven acidosis.
Atot	Albumin 3.5	Mildly low - slightly alkalinising.
Unmeasured anions	$AG\ 6 - 1 = 5$	Normal.
Mechanism	-	Chloride-driven NAGMA.

Where groups go wrong

- Expecting a raised anion gap because the child is dehydrated.
- Not pairing with Case 4 as a contrast: correction reveals nothing hidden here.

Your synthesis (final 2 min)

Pair explicitly with Case 4 - both have normal-looking raw AG and mild hypoalbuminaemia, but correction reveals nothing hidden here, whereas in Case 4 it unmasks a concealed HAGMA.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. What is the likely cause of the metabolic acidosis in this child?
- Q2. How do the chloride and lactate values help you identify the mechanism?

Answers

- A1. Loss of bicarbonate-rich fluid in large-volume diarrhoea $\rightarrow$ hyperchloraemic NAGMA.
- A2. Chloride elevated (112) as it replaces lost bicarbonate; normal lactate (1) argues against significant hypoperfusion/shock.

Post-intervention reveal

Intervention. IV isotonic fluids, correct dehydration.

pH	PaCO ₂	HCO ₃	Na	K
7.35	36	20	135	4.0
Cl	Albumin	Lactate		
105	3.5	1		

Step	Answer
6. Anion gap	AG = 135 - (105 + 20) = 10 - normal.
9. SID / Stewart	SID = 135 - 105 = 30 (improving).

Moderator take-home (final 1 min)

A normal-anion-gap acidosis with a normal lactate points to GI bicarbonate loss, not hypoperfusion. SID confirms the chloride-driven mechanism.

Case 4 - Suspected IEM

Small-group phase (8 min) - you run this

Vignette. 8-month-old girl with 1 day of fever and vomiting. Effortless tachypnoea, haemodynamically stable, lethargic. History of similar episodes in the past.

Presenting gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.24	38	16	38	135
K	Cl	Albumin	Glucose	Lactate
4	107	2	85	3

Worked interpretation

Step	Answer
1. Clinical history	Short acute illness in an infant with history of similar prior episodes - recurrence is the key word.
2. Internal validity	$[H+] = 24 \times 38 / 16 = 57 \rightarrow$ pH 7.24 - matches.
3. pH	7.24 - acidaemia.
4. Primary disorder	HCO ₃ 16 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 16 + 8 = 32 \pm 2$ (30-34). Actual 38 - higher than expected $\rightarrow$ additional respiratory acidosis.
6. Anion gap	Raw AG = $135 - (107 + 16) = 12$ (apparently normal). Albumin 2.0: corrected AG = $12 + 2.5 \times (4 - 2) = 17$ (clearly raised).
7. Delta ratio	Using corrected AG: $(17 - 12) / (24 - 16) = 5/8 = 0.625 \rightarrow$ mixed HAGMA + NAGMA.
8. Oxygenation & integration	Concealed HAGMA unmasked only after albumin correction, with inadequate respiratory compensation, in child with recurrent episodes $\rightarrow$ suspect IEM.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$135 - 107 = 28$	Low - chloride-driven component.
Atot	Albumin 2.0	Markedly low - alkalinising, masking HAGMA.
Unmeasured anions	$17 - 3 = 14$	Raised.
Mechanism	-	Three forces: low SID, low Atot (masking), unmeasured anions.

Where groups go wrong

- Stopping at AG 12, normal, without applying the albumin correction - the single most consequential miss in the whole case set.
- Treating lactate 3 as fully explaining the anion-gap elevation.

Your synthesis (final 2 min)

Correct the anion gap for albumin as a reflex, every time, especially when the raw number looks reassuring. Here it changes the entire diagnostic direction.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. Calculate the albumin-corrected anion gap.
- Q2. Is there an additional respiratory disorder?
- Q3. What is the significance of recurrent similar episodes in this clinical setting?

Answers

A1. $12 + 2.5 \times (4.0 - 2.0) = 17$ mEq/L. Raw gap of 12 looked normal; low albumin was masking a real HAGMA.

A2. Yes - Winter's predicts PaCO₂ 30-34; actual 38 is higher -> inadequate compensation (superimposed respiratory acidosis).

A3. Recurrent metabolic decompensation triggered by intercurrent illness raises strong suspicion for underlying IEM (organic acidaemia or fatty acid oxidation defect). Prompt metabolic work-up: ammonia, lactate, urine organic acids, plasma amino acids, acylcarnitine - the critical sample.

Post-intervention reveal

Intervention. Critical sample, dextrose, stop protein.

pH	PaCO ₂	HCO ₃	Na	K
7.35	35	20	138	4.2
Cl	Albumin	Glucose	Lactate	
105	2.5	100	2	

Step	Answer
9. SID / Stewart	SID = 138 - 105 = 33 (normalising). Atot still low (albumin 2.5) - still masking. Unmeasured anions = corrected AG ($20 + 2.5 \times 1.5 = 23.75$) - 2 = 21.75 (still raised).

Moderator take-home (final 1 min)

Correct the anion gap for albumin as a reflex, every time - it is most consequential exactly when the raw number looks reassuring. Also check SID.

Case 5 - Salicylate poisoning

Small-group phase (8 min) - you run this

Vignette. 15-year-old boy with tinnitus, fast breathing, vomiting, and confusion after intentional ingestion.

Presenting gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.42	18	11	108	140
K	Cl	Albumin	Glucose	Lactate
4.2	102	4.1	90	3.2

Worked interpretation

Step	Answer
1. Clinical history	Tinnitus, hyperventilation, vomiting, confusion after intentional ingestion in a teenager - tinnitus is the most specific clue.
2. Internal validity	$[H^+] = 24 \times 18 / 11 = 39 \rightarrow$ pH 7.41 - matches 7.42. Plausible arterial sample.
3. pH	7.42 - essentially normal (the trap).
4. Primary disorder	Two markedly abnormal primary values (PaCO ₂ 18, HCO ₃ 11) with near-normal pH - mixed disorder.
5. Expected compensation	If HCO ₃ 11 were the only problem, Winter's predicts PaCO ₂ 24.5 +/-2. Actual PaCO ₂ 18 is far below that - separate primary respiratory alkalosis.
6. Anion gap	AG = 140 - (102 + 11) = 27 (markedly raised). Lactate only partly explains.
7. Delta ratio	$(27 - 12)/(24 - 11) = 1.15 \rightarrow$ HAGMA.
8. Oxygenation & integration	Primary respiratory alkalosis + primary HAGMA, opposing pH effects cancelling to near-normal pH - classic salicylate poisoning.

SID / Stewart analysis

Tool	Value	Interpretation
SID	140 - 102 = 38	High-normal - no chloride-driven component.
Atot	Albumin 4.1	Normal.
Unmeasured anions	27 - 3.2 = 23.8	Markedly raised - salicylate metabolites + ketones.
Mechanism	-	Unmeasured anions + primary respiratory alkalosis.

Where groups go wrong

- Seeing pH 7.42 and concluding the gas is essentially normal.
- Attributing low PaCO₂ purely to appropriate compensation without checking against Winter's.

Your synthesis (final 2 min)

This case exists to break the habit of reading the pH first and stopping. Always check PaCO₂ and HCO₃ against each other.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. A near-normal pH can conceal a mixed disorder. What disorders are present here?
- Q2. What poisoning is suggested by the clinical history and ABG pattern?

Answers

A1. Primary respiratory alkalosis (PaCO₂ far lower than metabolic compensation alone would predict - direct salicylate stimulation of central respiratory centre) + primary HAGMA (AG 27, from lactate, ketosis, salicylate organic acids).

A2. Salicylate (aspirin) poisoning - classic triad of tinnitus, hyperventilation, and mixed respiratory-alkalosis/HAGMA pattern after intentional ingestion in an adolescent.

Post-intervention reveal

Intervention. Activated charcoal, IV fluids, sodium bicarbonate, alkalinise urine, consider dialysis.

pH	PaCO2	HCO3	Na	K
7.48	30	22	142	3.8
Cl	Albumin	Lactate		
105	4.1	1.5		

Step	Answer
9. SID / Stewart	SID = 142 - 105 = 37 (normal). Unmeasured anions = AG (142-105-22 = 15) - 1.5 = 13.5 (improving).

Moderator take-home (final 1 min)

A near-normal pH never means a normal gas - check PaCO2 and HCO3 against each other before you trust it. SID helps distinguish chloride-driven from unmeasured-anion-driven acidosis.

Case 6 - Post-arrest hanging

Small-group phase (8 min) - you run this

Vignette. 5-year-old boy, 20 kg, accidental hanging. Post-arrest, ROSC after 15 min CPR, ventilated. PEEP 6, PIP 16, RR 20, I:E 1:2, FiO2 100%, TV 80 mL, MAP 17.

Presenting gas

pH	PaCO2	HCO3	PaO2	Na
6.88	65	12	70	138
K	Cl	Albumin	Glucose	Lactate
6.2	98	4.1	210	3.2

Worked interpretation

Step	Answer
1. Clinical history	Post-cardiac-arrest after prolonged CPR for hanging - expect hypoxic-ischaemic metabolic acidosis and possible inadequate ventilation.
2. Internal validity	$[H^+] = 24 \times 65 / 12 = 130 \rightarrow \text{pH } 6.89$ - matches 6.88.
3. pH	6.88 - profound, life-threatening acidaemia.
4. Primary disorder	Both HCO3 12 and PaCO2 65 abnormal in acidifying direction - combined metabolic + respiratory acidosis.
5. Expected compensation	If pure metabolic, Winter's predicts PaCO2 26 +/-2. Actual 65 far above - large separate primary respiratory acidosis.
6. Anion gap	$AG = 138 - (98 + 12) = 28$ (markedly raised). Lactate 3.2 does not fully explain.
7. Delta ratio	$(28 - 12) / (24 - 12) = 1.33 \rightarrow$ HAGMA component.
8. Oxygenation & integration	$P/F = 70/1.0 = 70$. $OI = (17 \times 1.0 \times 100)/70 = 24.3$. Severe ARDS (PALICC OI ≥ 16).

SID / Stewart analysis

Tool	Value	Interpretation
SID	$138 - 98 = 40$	High-normal - no chloride-driven component.
Atot	Albumin 4.1	Normal.
Unmeasured anions	$28 - 3.2 = 24.8$	Markedly raised - hypoxic-ischaemic anions.
Mechanism	-	Unmeasured anions + primary respiratory acidosis.

Where groups go wrong

- Assuming high PaCO2 is simply the ventilator not yet caught up without naming it as a second primary disorder.
- Not flagging K 6.2 as immediate actionable safety issue.
- Not noticing lactate 3.2 under-explains AG 28.

Your synthesis (final 2 min)

Most acidotic gas in the set exists to force explicit separation of the two acidoses and bring in oxygenation indices.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. Identify all components of the mixed acid-base disorder.
- Q2. Calculate the P/F ratio and oxygenation index using the available data.
- Q3. How would you grade the severity of oxygenation failure?

Answers

A1. (1) Primary metabolic acidosis (HCO3 12, AG 28 - hypoxic-ischaemic/lactic) and (2) primary respiratory acidosis (PaCO2 65, far exceeding expected ~26) - combined metabolic + respiratory acidosis.

A2. $P/F = 70/1.0 = 70$. $OI = (17 \times 1.0 \times 100)/70 = 24.3$.

A3. $OI\ 24.3$ = severe oxygenation failure (PALICC $OI \geq 16$ = severe).

Post-intervention reveal

Intervention. Increase ventilation, treat hyperkalaemia, fluids, consider bicarbonate.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.25	40	17	90	138
K	Cl	Albumin	Glucose	Lactate
4.5	100	4.1	180	2.5

Step	Answer
9. SID / Stewart	$SID = 138 - 100 = 38$ (normal). Unmeasured anions = $AG\ (138 - 100 - 17 = 21) - 2.5 = 18.5$ (still raised).

Moderator take-home (final 1 min)

When PaCO₂ and HCO₃ are both markedly abnormal in the same direction, name both disorders explicitly - one is not compensating for the other. SID confirms no chloride-driven component.

Case 7 - Septic shock + severe anaemia

Small-group phase (8 min) - you run this

Vignette. 4-year-old boy with septic shock and severe anaemia. FiO₂ 40%, SpO₂ 99%, ScvO₂ 42%.

Presenting gas

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.25	26	11	110	138
K	Cl	Albumin	Hb	Lactate
4.3	106	4.1	4.5	7

Worked interpretation

Step	Answer
1. Clinical history	Septic shock with severe anaemia (Hb 4.5) - two independent reasons for impaired oxygen delivery.
2. Internal validity	$[H^+] = 24 \times 26 / 11 = 57 \rightarrow$ pH 7.25 - matches.
3. pH	7.25 - acidaemia.
4. Primary disorder	HCO ₃ 11 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 11 + 8 = 24.5 \pm 2$. Actual 26 - appropriate.
6. Anion gap	$AG = 138 - (106 + 11) = 21$ (raised). Albumin normal, negligible correction.
7. Delta ratio	$(21 - 12) / (24 - 11) = 0.69 \rightarrow$ mixed HAGMA + NAGMA.
8. Oxygenation & integration	$P/F = 110 / 0.4 = 275$. ScvO ₂ 42% - tissues extracting far more O ₂ than usual. $CaO_2 = (1.34 \times 4.5 \times 0.99) + (0.003 \times 110) = 6.3$ mL/dL.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$138 - 106 = 32$	Low - chloride-driven component from resuscitation fluids.
Atot	Albumin 4.1	Normal.
Unmeasured anions	$21 - 7 = 14$	Raised - lactic acidosis.
Mechanism	-	Two forces: low SID + unmeasured anions.

Where groups go wrong

- Reassurance from normal SpO₂/PaO₂ without looking at ScvO₂.
- Forgetting that severe anaemia is, by itself, full explanation for impaired oxygen delivery.

Your synthesis (final 2 min)

Anchor on oxygen-delivery equation ($DO_2 = \text{cardiac output} \times CaO_2$, and CaO_2 driven mainly by $Hb \times SO_2$).

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. Calculate the P/F ratio.
- Q2. How do you interpret the low ScvO₂ despite a normal SpO₂?
- Q3. What do the severe anaemia and elevated lactate indicate about oxygen delivery and extraction?

Answers

- A1. $110 / 0.4 = 275$ - mild oxygenation impairment; not where the problem lies.
- A2. Large veno-arterial oxygen extraction gap: lungs saturating haemoglobin normally, but tissues extracting abnormally high fraction because overall DO_2 inadequate relative to demand. SpO₂ alone misses this.

A3. Severe anaemia critically reduces CaO₂ and therefore DO₂, even though saturation normal. Body compensates initially by increasing extraction (low ScvO₂), but once exceeded, anaerobic metabolism follows - lactate ↑. Content/delivery problem, not lung problem. Priority: transfusion.

Post-intervention reveal

Intervention. Blood transfusion, antibiotics, fluids, vasopressors.

pH	PaCO ₂	HCO ₃	PaO ₂	Na
7.35	36	20	100	138
K	Cl	Albumin	Hb	Lactate
4.0	105	3.8	9.0	2.5
ScvO ₂				
65%				

Step	Answer
9. SID / Stewart	SID = 138 - 105 = 33 (low-normal). Unmeasured anions = AG (138-105-20 = 13) - 2.5 = 10.5 (normalising).

Moderator take-home (final 1 min)

A normal SpO₂ says nothing about oxygen delivery when haemoglobin is this low - look at ScvO₂ and lactate, not the saturation monitor. SID reveals the chloride-driven component.

Case 8 - Methaemoglobinaemia

Small-group phase (8 min) - you run this

Vignette. 10-year-old boy with cyanosis, headache, tachycardia, and lethargy after receiving topical benzocaine. Blood sample drawn looks chocolate-brown. FiO2 90%, SpO2 85%, SO2 65%.

Presenting gas

pH	PaCO2	HCO3	PaO2	Na
7.36	38	21	550	138
K	Cl	Albumin	MetHb	Lactate
4.3	105	4.1	35%	2

Worked interpretation

Step	Answer
1. Clinical history	Cyanosis, headache, tachycardia, lethargy after topical benzocaine, with chocolate-brown blood - close to diagnostic before numbers read.
2. Internal validity	$[H^+] = 24 \times 38 / 21 = 43 \rightarrow$ pH 7.36 - matches. Essentially normal acid-base gas.
3. pH	7.36 - essentially normal.
4. Primary disorder	No significant primary acid-base disorder. Acid-base is distraction.
5. Expected compensation	Not applicable.
6. Anion gap	$AG = 138 - (105 + 21) = 12$ (normal).
7. Delta ratio	Not applicable.
8. Oxygenation & integration	$P/F = 550/0.9 = 611$ (remarkably high). Yet SpO2 85%, SO2 65% - large discordant saturation gap. MetHb 35% (normal <1-2%).

SID / Stewart analysis

Tool	Value	Interpretation
SID	$138 - 105 = 33$	Low-normal - mild chloride-driven component.
Atot	Albumin 4.1	Normal.
Unmeasured anions	$12 - 2 = 10$	Normal. Acid-base status benign.
Mechanism	-	Acid-base is not the story; oxygenation is.

Where groups go wrong

- Getting drawn into acid-base analysis when the gas is essentially normal.
- Assuming a high PaO2 guarantees adequate oxygen delivery.

Your synthesis (final 2 min)

Close with the rule this case exists to teach: a normal or high PaO2 does not guarantee normal oxygen-carrying capacity.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. Calculate the P/F ratio.
- Q2. How do you explain the discrepancy between the very high PaO2, SpO2, and SO2?
- Q3. What is the significance of the chocolate-brown blood and MetHb 35%?
- Q4. What is the most likely diagnosis?

Answers

A1. $550 / 0.9 \sim 611$.

A2. PaO2 measures dissolved oxygen tension and is unaffected by methaemoglobin, so remains high. Methaemoglobin (Fe3+, unable to bind oxygen) distorts pulse oximeter algorithm, driving SpO2 toward falsely fixed value ~82-85%. Directly measured SO2 by co-oximetry (65%) reflects true, much lower functional saturation - 35% of haemoglobin cannot carry oxygen at all.

A3. Chocolate-brown blood is classic bedside sign. MetHb 35% confirms severe elevation - enough to cause significant functional anaemia (only 65% of haemoglobin can carry oxygen). Explains cyanosis, headache, tachycardia, lethargy despite high PaO2.

A4. Acquired methaemoglobinaemia (benzocaine-induced).

Post-intervention reveal

Intervention. Methylene blue 1-2 mg/kg IV (check G6PD status first), oxygen, stop benzocaine.

pH	PaCO2	HCO3	PaO2	Na
7.38	38	23	200	138
K	Cl	Albumin	MetHb	Lactate
4.0	105	4.1	5%	1.5
SpO2	SO2			
98%	95%			

Step	Answer
9. SID / Stewart	SID = 138 - 105 = 33 (low-normal). Unmeasured anions = AG (138-105-23 = 10) - 1.5 = 8.5 (normal).

Moderator take-home (final 1 min)

A high PaO2 does not guarantee oxygen-carrying capacity - whenever SpO2 and PaO2 disagree, request co-oximetry before you do anything else.

Case 9 - Two encephalopathic neonates

Small-group phase (8 min) - you run this

Vignette. Baby A: 4 days old, fed well until day 2, now lethargic and tachypnoeic. Afebrile, clear chest. On antibiotics for sepsis. Baby B: 7 days old, vomiting, lethargic, deep rapid breathing. Platelets 60,000/uL, urine ketones 3+.

Presenting gas

A pH	A PaCO ₂	A HCO ₃	A SBE	A Na/Cl
7.51	26	20	-1	138/104
A Glucose	A Lactate	B pH	B PaCO ₂	B HCO ₃
85	1.8	7.14	18	6
B SBE	B Na/Cl	B Glucose	B Lactate	
-21	136/100	45	3.0	

Worked interpretation

Step	Answer
1. Clinical history	A: fed well until day 2, now lethargic and tachypnoeic; sepsis label but afebrile, clear chest. B: vomiting, lethargic, deep rapid breathing, thrombocytopenia, ketonuria.
2. Internal validity	A: $[H^+] = 24 \times 26 / 20 = 31 \rightarrow$ pH 7.51 - matches. B: $[H^+] = 24 \times 18 / 6 = 72 \rightarrow$ pH 7.14 - matches.
3. pH	A: 7.51 - alkalaemia. B: 7.14 - acidaemia.
4. Primary disorder	A: PaCO ₂ 26 (low) $\rightarrow$ respiratory alkalosis. B: HCO ₃ 6 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	A: acute resp alkalosis expected HCO ₃ $\sim$ 21.2; actual 20 - appropriate. B: Winter's predicted PaCO ₂ 17 ± 2 ; actual 18 - appropriate.
6. Anion gap	A: AG = $138 - (104 + 20) = 14$ (normal). B: AG = $136 - (100 + 6) = 30$ (markedly raised).
7. Delta ratio	A: not applicable. B: $(30 - 12)/(24 - 6) = 1.0 \rightarrow$ HAGMA.
8. Oxygenation & integration	A: sepsis alone does not explain respiratory alkalosis - suspect hyperammonaemia. B: lactate only 3.0 - rest of gap filled by ketones and organic acids.

SID / Stewart analysis

Tool	Value	Interpretation
A SID	$138 - 104 = 34$	Low-normal.
A Unmeasured	$14 - 1.8 = 12.2$	Normal. Mechanism: respiratory alkalosis.
B SID	$136 - 100 = 36$	Normal.
B Unmeasured	$30 - 3 = 27$	Markedly raised - organic acids/ketones.

Where groups go wrong

- Attributing Baby A alkalosis to sepsis without considering hyperammonaemia.
- Not checking ammonia on ice for both babies.
- Assuming lactate explains Baby B AG when it only accounts for part.

Your synthesis (final 2 min)

Both babies have a metabolic disorder masquerading as sepsis. The gas is the clue - ammonia on ice is the test.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. Name the primary disorder in each baby. Is compensation appropriate in each?
- Q2. Which features does sepsis alone fail to explain?

Q3. Baby B has an anion gap of 30 but lactate is only 3. What fills the rest?

Q4. Intervene: what one test do you send in both, how must the sample be handled, and what do you do while waiting?

Answers

A1. A: respiratory alkalosis, appropriate. B: metabolic acidosis, appropriate.

A2. A: respiratory alkalosis without fever or distress. B: severe HAGMA with ketones and hypoglycaemia.

A3. Ketones (urine 3+) and organic acids from inborn error of metabolism.

A4. Ammonia. Sample on ice, delivered to lab within 15 minutes. While waiting: stop protein, give dextrose, prepare ammonia scavengers.

Post-intervention reveal

Intervention. Both: ammonia level, stop protein, dextrose. A: ammonia scavengers if high. B: carnitine, treat acidosis.

A pH	A PaCO ₂	A HCO ₃	A Na/Cl	A Glucose
7.45	35	22	138/105	90
A Lactate	B pH	B PaCO ₂	B HCO ₃	B Na/Cl
1.5	7.30	25	12	138/102
B Glucose	B Lactate			
100	2.0			

Step	Answer
9. SID / Stewart	A: SID = 138 - 105 = 33; unmeasured anions normal. B: SID = 138 - 102 = 36; unmeasured anions = AG (138-102-12 = 24) - 2 = 22 (still raised).

Moderator take-home (final 1 min)

Neonatal encephalopathy + abnormal gas = metabolic screen; ammonia on ice is critical. SID helps distinguish chloride-driven from unmeasured-anion-driven acidosis.

Case 10 - HUS-like

Small-group phase (8 min) - you run this

Vignette. 3-year-old, 5 days after bloody diarrhoea, now oliguric and pale. Hb 6.2 g/dL, platelets 40,000/uL, creatinine 3.8 mg/dL.

Presenting gas

pH	PaCO ₂	HCO ₃	SBE	Na
7.22	25	10	-16	131
K	Cl	Albumin	Phosphate	iCa
6.8	104	3.0	9.5	0.92

Worked interpretation

Step	Answer
1. Clinical history	Bloody diarrhoea followed by oliguria and pallor, with anaemia, thrombocytopenia, and renal failure - HUS-like picture.
2. Internal validity	$[H+] = 24 \times 25 / 10 = 60 \rightarrow$ pH 7.22 - matches.
3. pH	7.22 - acidaemia.
4. Primary disorder	HCO ₃ 10 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 10 + 8 = 23 \pm 2$. Actual 25 - appropriate.
6. Anion gap	$AG = 131 - (104 + 10) = 17$. Albumin-corrected $AG = 17 + 2.5 \times (4 - 3.0) = 19.5$ (raised).
7. Delta ratio	$(17 - 12) / (24 - 10) = 5/14 = 0.36 \rightarrow$ mixed NAGMA + HAGMA.
8. Oxygenation & integration	Mixed acidosis; hyperkalaemia and hypocalcaemia are immediate threats.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$131 - 104 = 27$	Markedly low - chloride-driven.
Atot	Albumin 3.0 (alkalinising), phosphate 9.5 (markedly acidifying)	Net high Atot.
Unmeasured anions	$19.5 - 0 = 19.5$	Raised - urate, oxalate, retained organic anions.
Mechanism	-	Three forces: low SID + high Atot (phosphate) + unmeasured anions.

Where groups go wrong

- Missing that all three Stewart forces are active here.
- Not recognising K 6.8 and iCa 0.92 as immediate actionable threats.
- Reaching for bicarbonate without considering the risks.

Your synthesis (final 2 min)

This is the triple-hit acidosis case. SID, Atot, and unmeasured anions each contribute. Treat the electrolytes first.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1.** Is compensation appropriate? What are the anion gap and the albumin-corrected anion gap?
- Q2.** Calculate the delta ratio. What does it tell you?
- Q3.** Using SID and Atot, name the three separate forces acidifying this child.
- Q4.** Intervene: which two sidekick values change your next 10 minutes? The team asks for bicarbonate - what do you say?

Answers

A1. Appropriate. AG 17, corrected AG 19.5.

A2. 0.36 -> mixed NAGMA + HAGMA.

A3. Low SID (chloride-driven), high Atot (phosphate), unmeasured anions.

A4. K 6.8 (hyperkalaemia) and iCa 0.92 (hypocalcaemia). Bicarbonate? No - pH 7.22 not severe; bicarbonate can worsen hypocalcaemia and hyperkalaemia.

Post-intervention reveal

Intervention. Dialysis, treat hyperkalaemia and hypocalcaemia.

pH	PaCO ₂	HCO ₃	Na	K
7.35	35	18	135	4.5
Cl	Albumin	Phosphate	iCa	
105	3.2	6.0	1.0	

Step	Answer
9. SID / Stewart	SID = 135 - 105 = 30 (improving). Atot: phosphate improved from 9.5 to 6.0. Unmeasured anions = AG (135-105-18 = 12) - 0 = 12 (normalising).

Moderator take-home (final 1 min)

HUS = mixed acidosis; hyperkalaemia and hypocalcaemia are immediate threats. SID, Atot, and unmeasured anions each tell a different part of the story.

Case 11 - Hepatitis A encephalopathy

Small-group phase (8 min) - you run this

Vignette. 6-year-old with hepatitis A, grade II encephalopathy. INR 4.2, ammonia 180 umol/L.

Presenting gas

pH	PaCO ₂	HCO ₃	SBE	Na
7.49	28	21	-1	132
K	Cl	Albumin	Lactate	Glucose
3.0	98	2.2	5.5	45

Worked interpretation

Step	Answer
1. Clinical history	Hepatitis A with grade II encephalopathy, coagulopathy, hyperammonaemia.
2. Internal validity	$[H^+] = 24 \times 28 / 21 = 32 \rightarrow$ pH 7.49 - matches.
3. pH	7.49 - alkalaemia.
4. Primary disorder	PaCO ₂ 28 (low) $\rightarrow$ respiratory alkalosis.
5. Expected compensation	Acute resp alkalosis: HCO ₃ falls ~2 per 10 mmHg fall in PaCO ₂ . PaCO ₂ 28 is 12 below 40 $\rightarrow$ expected HCO ₃ ~21.6. Actual 21 - appropriate.
6. Anion gap	$AG = 132 - (98 + 21) = 13$. Corrected AG = $13 + 2.5 \times (4 - 2.2) = 17.5$ (HAGMA masked by hypoalbuminaemia).
7. Delta ratio	Using corrected AG: $(17.5 - 12)/(24 - 21) = 5.5/3 = 1.83 \rightarrow$ mixed.
8. Oxygenation & integration	Respiratory alkalosis + masked HAGMA + hypoglycaemia.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$132 - 98 = 34$	Low-normal.
Atot	Albumin 2.2	Markedly low - alkalinising, masking HAGMA.
Unmeasured anions	$17.5 - 5.5 = 12$	Raised.
Mechanism	-	Masked HAGMA + respiratory alkalosis.

Where groups go wrong

- Taking SBE -1 at face value without correcting AG for albumin.
- Missing that the low albumin is masking a HAGMA.
- Not treating hypoglycaemia urgently.

Your synthesis (final 2 min)

The SBE looks normal, but the low albumin is lying to you. Correct the AG, check SID, and treat the glucose.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1.** What is the primary disorder, and why is this child alkaline?
- Q2.** Calculate the anion gap, then correct it for albumin. What changes?
- Q3.** The base excess is -1, which looks normal. Why is that misleading? How many processes are running at once?
- Q4.** Intervene: which two sidekicks need treating in the next few minutes, and why does one of them worsen the encephalopathy?

Answers

- A1.** Respiratory alkalosis due to hyperventilation from ammonia stimulating respiratory centre.

A2. AG 13, corrected AG 17.5. Low albumin was masking a HAGMA.

A3. Yes - because HAGMA is masked by hypoalbuminaemia. Multiple processes: respiratory alkalosis, HAGMA (lactic acidosis), hypoglycaemia.

A4. Glucose 45 (hypoglycaemia) and lactate 5.5 (lactic acidosis). Hypoglycaemia worsens encephalopathy.

Post-intervention reveal

Intervention. Dextrose, lactulose, ammonia scavengers.

pH	PaCO ₂	HCO ₃	Na	K
7.42	32	20	135	3.5
Cl	Albumin	Lactate	Glucose	
100	2.5	3.0	100	

Step	Answer
9. SID / Stewart	SID = 135 - 100 = 35 (normal). Atot still low (albumin 2.5) - still masking. Unmeasured anions = corrected AG (15 + 3.75 = 18.75) - 3 = 15.75 (still raised).

Moderator take-home (final 1 min)

Hepatitis A encephalopathy = respiratory alkalosis + HAGMA; check glucose. Always correct the AG for albumin, and check SID - the SBE may look normal while a HAGMA is masked.

Case 12 - Fasting hypoglycaemia

Small-group phase (8 min) - you run this

Vignette. 18-month-old, previously well, fasted overnight after gastroenteritis and presented with a seizure. Liver palpable 3 cm. ALT 180 U/L, ammonia 110 umol/L, free fatty acids raised.

Presenting gas

Glucose	pH	PaCO ₂	HCO ₃	SBE
30	7.36	36	20	-4
Na/Cl 138/105	Lactate 2.0	b-OHB 0.3		

Worked interpretation

Step	Answer
1. Clinical history	Fasting hypoglycaemia with seizure, hepatomegaly, raised ALT and ammonia, raised free fatty acids - key is the inappropriately low ketones.
2. Internal validity	$[H+] = 24 \times 36 / 20 = 43 \rightarrow$ pH 7.36 - matches.
3. pH	7.36 - normal.
4. Primary disorder	HCO ₃ 20 (slightly low), SBE -4 $\rightarrow$ mild metabolic acidosis.
5. Expected compensation	Not applicable (primary metabolic).
6. Anion gap	$AG = 138 - (105 + 20) = 13$ (normal).
7. Delta ratio	Not applicable (NAGMA).
8. Oxygenation & integration	Near-normal gas but hypoglycaemia with low ketones $\rightarrow$ failure of ketogenesis.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$138 - 105 = 33$	Low-normal - mild chloride-driven component.
Atot	Albumin not given	-
Unmeasured anions	$13 - 2 = 11$	Normal.
Mechanism	-	Acidosis mild and chloride-driven; the story is ketogenesis failure.

Where groups go wrong

- Being reassured by the near-normal gas and missing the hypoglycaemia with inappropriately low ketones.
- Drawing the critical sample after giving dextrose.

Your synthesis (final 2 min)

The gas is a distraction. The diagnosis is in the glucose and ketones. Critical sample before dextrose.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. The gas looks almost normal. Why is that the most worrying finding?
- Q2. What should a fasting child gas and ketones look like?
- Q3. Free fatty acids are high but ketones are low. What does that mean? How would hyperinsulinism differ?
- Q4. Intervene: list the critical sample. When exactly must it be drawn, and what do you give immediately afterwards?

Answers

- A1. In a fasting child, expected ketosis and metabolic acidosis. Normal gas with low b-OHB (0.3) indicates failure of ketogenesis.
- A2. Metabolic acidosis with high anion gap due to ketones.

- A3.** Block in fatty acid oxidation or ketogenesis (e.g., MCAD deficiency). Hyperinsulinism would have low FFA and low ketones.
- A4.** Must be drawn during hypoglycaemia before treatment. Includes glucose, insulin, C-peptide, cortisol, growth hormone, ketones, free fatty acids, acylcarnitine profile, ammonia, lactate, LFTs. Give dextrose (2-3 mL/kg of 10% dextrose) immediately after.

Post-intervention reveal

Intervention. Critical sample, then dextrose.

Glucose	pH	PaCO2	HCO3	SBE
100	7.38	38	22	-1
Na/Cl	Lactate	b-OHB		
138/105	1.5	0.2		

Step	Answer
9. SID / Stewart	SID = 138 - 105 = 33 (low-normal). Unmeasured anions = AG (138-105-22 = 11) - 1.5 = 9.5 (normal).

Moderator take-home (final 1 min)

Fasting hypoglycaemia with low ketones = think fatty acid oxidation defect; critical sample before treatment. SID confirms the acidosis was mild and chloride-driven, not the primary problem.

Case 13 - Shocked 2-week-old boy

Small-group phase (8 min) - you run this

Vignette. 2-week-old boy with 3 days of vomiting and poor feeding, now mottled and lethargic. Genitalia look normal. Weight below birth weight.

Presenting gas

pH	PaCO ₂	HCO ₃	SBE	Na
7.27	28	12.5	-13	122
K	Cl	Glucose		
7.8	98	40		

Worked interpretation

Step	Answer
1. Clinical history	Neonatal shock with vomiting, poor feeding, mottling, lethargy, weight loss.
2. Internal validity	$[H^+] = 24 \times 28 / 12.5 = 54 \rightarrow$ pH 7.27 - matches.
3. pH	7.27 - acidaemia.
4. Primary disorder	HCO ₃ 12.5 (low) $\rightarrow$ metabolic acidosis.
5. Expected compensation	Winter's: $PaCO_2 = 1.5 \times 12.5 + 8 = 26.75 \pm 2$. Actual 28 - appropriate.
6. Anion gap	$AG = 122 - (98 + 12.5) = 11.5$ (normal). NAGMA.
7. Delta ratio	Not applicable.
8. Oxygenation & integration	NAGMA with hyponatraemia, hyperkalaemia, hypoglycaemia $\rightarrow$ CAH salt-wasting crisis.

SID / Stewart analysis

Tool	Value	Interpretation
SID	$122 - 98 = 24$	Markedly low - chloride-driven acidosis.
Atot	Albumin not given	-
Unmeasured anions	$11.5 - 0 = 11.5$	Normal.
Mechanism	-	Chloride-driven NAGMA.

Where groups go wrong

- Not recognising the triad of hyponatraemia, hyperkalaemia, and hypoglycaemia as CAH.
- Giving hydrocortisone before drawing the critical sample.
- Giving potassium-containing fluids to a hyperkalaemic neonate.

Your synthesis (final 2 min)

Neonatal shock + NAGMA + hyperkalaemia + hyponatraemia = CAH until proven otherwise. Draw the sample before steroids.

Plenary phase (7 min) - moderator runs this

Case-specific questions

- Q1. Classify the acid-base disorder and check compensation. Is this a HAGMA or a NAGMA?
- Q2. Three sidekick values make the diagnosis before any specialised test. Which, and what is it?
- Q3. A vomiting 2-week-old with pyloric stenosis would show which gas instead?
- Q4. Intervene: what are your first three actions, and which sample must precede them?

Answers

- A1. NAGMA with appropriate compensation.
- A2. Na 122 (hyponatraemia), K 7.8 (hyperkalaemia), glucose 40 (hypoglycaemia) $\rightarrow$ CAH.

A3. Metabolic alkalosis (hypochloraemic, hypokalaemic).

A4. (1) IV access, draw critical sample (electrolytes, glucose, cortisol, ACTH, 17-OHP, renin, aldosterone). (2) Give stress-dose hydrocortisone. (3) IV fluids with dextrose and sodium (normal saline bolus, then D5NS; no potassium initially due to hyperkalaemia). Sample must precede hydrocortisone.

Post-intervention reveal

Intervention. Hydrocortisone, fluids with dextrose, treat hyperkalaemia.

pH	PaCO2	HCO3	SBE	Na
7.35	35	18	-5	135
K	Cl	Glucose		
5.0	100	90		

Step	Answer
9. SID / Stewart	SID = 135 - 100 = 35 (normalising). Unmeasured anions = AG (135-100-18 = 17) - 0 = 17 (raised - resolving).

Moderator take-home (final 1 min)

Neonatal shock + NAGMA + hyperkalaemia + hyponatraemia = CAH until proven otherwise. SID confirms the chloride-driven mechanism and tracks response to therapy.